E-In Son

Seoul, Republic of Korea · pingpang@snu.ac.kr · Homepage · LinkedIn

Research Interests

My research focuses on learning-based navigation for robots in real-world environments, spanning traversability estimation in unstructured terrain, visual navigation, and world models. Going forward, I aim to develop world models and navigation policies that generalize across environments and robot embodiments.

Education

Seoul National University **Mar 2021 – Aug 2027 (Expected)**
Ph.D. Student, Dept. of Electrical & Computer Engineering *Seoul, Korea*
Vehicle Intelligence Laboratory. (Advisor: Prof. Seung-Woo Seo)

Konkuk University **Mar 2015 – Feb 2021**
B.S., Electrical & Electronics Engineering *Seoul, Korea*
Summa Cum Laude (GPA 4.33 / 4.5).

Publications

[1] **Feeling Terrain Before Crossing: World Models for Off-Road Navigation.**
E-In Son, Dong-Wook Kim, Ji-Hoon Hwang, Kangsun Lee, Jisung Bae, Jung-Taak Kim, Seung-Woo Seo.
Under review, 2026. [Paper] [Code]

[2] **Which Terrain Is Better? Preference Learning with VLM Prototypes for Off-Road Traversability Ranking.**
Ji-Hoon Hwang, Jisung Bae, E-In Son, Dong-Wook Kim, Jung-Taak Kim, Seung-Woo Seo.
Under review, 2026. [Paper]

[3] **CanonNav: Disentangling Navigation Behavior from Camera Geometry in Cross-Platform Visual Navigation.**
Dong-Wook Kim, Ji-Hoon Hwang, E-In Son, Mintaek Oh, Seung-Woo Seo.
Under review, 2026. [Paper]

[4] **Ordinal Neural Collapse as a Representation Prior for Visual Navigation.**
E-In Son*, Jung-Taak Kim*, Seung-Woo Seo.
European Conference on Computer Vision (ECCV), 2026. [Paper] [Project]

[5] **PTS-Map: Probabilistic Terrain State Map for Uncertainty-Aware Traversability Mapping in Unstructured Environments.**
Dong-Wook Kim, E-In Son, Chan Kim, Ji-Hoon Hwang, Seung-Woo Seo.
IEEE Robotics and Automation Letters (RA-L), 2025. [Paper] [Code]

[6] **Follow the Footprints: Self-supervised Traversability Estimation for Off-road Vehicle Navigation based on Geometric and Visual Cues.**
Yurim Jeon, E-In Son, Seung-Woo Seo.
IEEE International Conference on Robotics and Automation (ICRA), 2024. [Paper] [Code]

[7] **Adaptive Robot Traversability Estimation Based on Self-Supervised Online Continual Learning in Unstructured Environments.**
Hyung-Suk Yoon, Ji-Hoon Hwang, Chan Kim, E-In Son, Se-Wook Yoo, Seung-Woo Seo.
IEEE Robotics and Automation Letters (RA-L), 2024. [Paper]

[8] **Traversability-aware Adaptive Optimization for Path Planning and Control in Mountainous Terrain.**
Se-Wook Yoo*, E-In Son*, Seung-Woo Seo.
IEEE Robotics and Automation Letters (RA-L), 2024. [Paper] [Project]

Awards & Honors

Earth Rover Challenge (1st Place among AI Teams) Oct 2024
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
Led planning for our team's navigation system, deployed remotely on sidewalk robots in 8 cities worldwide and benchmarked against human teleoperators.

Academic Excellence Scholarship 2018 – 2020
Konkuk University

Research Projects

Challengeable Future Defense Technology R&D Program (Phase 2) 2024 – 2026
Agency for Defense Development (ADD)

- Led planning for an end-to-end visual navigation system in unknown, hazardous terrain, building on our visual navigation methods [3, 4].
- Deployed on a Clearpath Husky and a custom tracked robot at 10 Hz, validated over 20 field runs of 1 km each.

Challengeable Future Defense Technology R&D Program (Phase 1) 2022 – 2023
Agency for Defense Development (ADD)

- Developed traversability-aware path planning and tracking for steep mountainous terrain, adapting to the robot's speed and torque; validated on 27 simulated terrain types and in real-world tests [8].

Teaching Experience

Seoul National University, Teaching Assistant

SNU in Silicon Valley Program, joint with UC Berkeley *(Summer 2024)*
Fundamentals of Random Variables and Probability Processes *(Spring 2023)*
Network Protocol and Design *(Fall 2021, 2023)*

Academic Service

Conference Reviewer: ICRA, IROS, ECCV
Journal Reviewer: RA-L

Skills

Programming	Python, C/C++, Git
Frameworks	PyTorch, ROS / ROS 2
Robotics	SLAM & localization, semantic segmentation, path planning & tracking, multi-sensor integration (LiDAR, camera, GNSS, IMU)
Simulation	Isaac Sim, MuJoCo, Gazebo, CARLA
Communication	Korean (native), English

Reference

Advisor: Prof. Seung-Woo Seo sseo@snu.ac.kr Website